Gears and wheels

antocciphysics | TEACHER KEY | CPA

Learning target: Use gear ratio and wheel circumference to predict an ideal travel speed.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 18; 3 course days later.*
15 J; heating or sound are possible.

02. Explain a worked example (worked)
Speed ratio driven/driver=10/30=1/3; directions opposite for an external pair.

03. Look and explain (model)
Larger driven gear turns more slowly; meshed external gears turn opposite ways.

04. Build the solution (practice)
2 rev/s; .40 m/s.

05. Use the idea (reason)
Larger wheels need more axle torque for the same driving force, may add inertia/load, and may slow the motor or slip.

06. Choose a method (mixed) *Review: Unit 7, Day 9; 12 course days later.*
7 A in each branch.

07. Exit check (exit)
Doubles ideal speed; actual rpm/load/slip can change.

08. STRETCH: Apply it another way (stretch)
Examples 10: 30 and 15: 45; need motor speed, wheel size, slip and load.

Gears and wheels

antocciphysics | TEACHER KEY | HONORS

Learning target: Use gear ratio and wheel circumference to predict an ideal travel speed.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 18; 3 course days later.*
$|\text{emf}| = N|\Delta\Phi|/\Delta t = 0.5$ V.

02. Explain a worked example (worked)
Speed ratio driven/driver=10/30=1/3; directions opposite for an external pair.

03. Represent and interpret (model)
Larger driven gear turns more slowly; meshed external gears turn opposite ways.

04. Solve and justify (practice)
200 rpm; C=.1885 m; speed=.628 m/s.

05. Analyze the reasoning (reason)
Larger wheels need more axle torque for the same driving force, may add inertia/load, and may slow the motor or slip.

06. Choose a method (mixed) *Review: Unit 7, Day 9; 12 course days later.*
Parallel 5 Ω; total 12 Ω; supply current 2 A.

07. Exit check (exit)
Torque ×3 and speed ×1/3, ignoring losses.

08. STRETCH: Apply it another way (stretch)
300 rpm=5 turns/s; $v = 2\pi(0.03)\times 5 \approx 0.942$ m/s.